

SRI SAIRAM COLLEGE OF ENGINEERING

BENGALURU

INDUSTRIAL INTERNSHIP REPORT

ON

PREDICTION OF AGRICULTURE CROP PRODUCTION IN INDIA

Submitted by

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Internship facilitated by upskill Campus and The IoT Academy
in association with UniConverge Technologies Pvt. Ltd. (UCT)

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Executive Summary

This report describes AgriPredict, a machine-learning-based agricultural crop production prediction application focused on Indian agricultural records. The system uses historical crop information to estimate production from user-selected agricultural inputs and presents the result through an interactive Streamlit web application.

The project workflow includes loading a CSV dataset, inspecting and preprocessing the data, preparing categorical and numerical features, training a Random Forest Regression model, evaluating the model, saving the trained model with Joblib, and using the saved model in the web application.

The supplied project material records a starter dataset of 60 rows and 7 columns. The reported evaluation results are MAE 108,681.79, RMSE 129,101.07 and R^2 0.447. Since the dataset is small, the system is a learning and demonstration prototype; its predictions should not be treated as verified nationwide agricultural forecasts.

Report at a Glance

Item	Details
Project title	Prediction of Agriculture Crop Production in India
Application name	AgriPredict – Smart Agriculture Platform
Student	Vijayalakshmi
Department / Institution	Computer Science and Engineering / Sri Sairam College of Engineering, Bengaluru
Internship partners	upskill Campus, The IoT Academy and UniConverge Technologies Pvt. Ltd. (UCT)
Domain	Data Science and Machine Learning
Core technologies	Python, Pandas, scikit-learn, Random Forest Regression, Streamlit and Joblib
Dataset	crop_production_starter_dataset.csv (60 records, 7 columns)

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1. Preface

This report summarizes the development of AgriPredict as an academic and internship project. The project explores how historical agricultural records can be used with machine learning to estimate crop production and how those estimates can be made accessible through a simple web interface.

The work provided practical exposure to data loading, data inspection, preprocessing, regression modelling, evaluation metrics, model persistence and application development. The report uses the project details and results supplied for this work and distinguishes recorded results from illustrative examples.

2. Introduction

2.1 Project Background

Agriculture is an important sector in India. Crop production varies with location, crop type, agricultural season, cultivation year and the area used for cultivation. Historical agricultural records can be analysed to understand production patterns and build data-driven estimates.

AgriPredict accepts agricultural information and uses a trained regression model to estimate crop production. The Streamlit interface provides a convenient way to explore input combinations and view the model output.

2.2 Internship Objectives

- Gain practical experience in an industry-oriented data science project.
- Understand how machine learning can be applied to an agricultural problem.
- Improve Python programming, data handling and analytical skills.
- Learn model training, evaluation and deployment concepts.
- Develop debugging, documentation and technical communication skills.

2.3 Project Objectives

- Collect and understand agricultural crop production data.
- Preprocess the dataset and prepare features for modelling.
- Identify the input features and target variable.
- Develop a regression model to estimate crop production.
- Evaluate the model using MAE, RMSE and R^2.
- Save the trained model and integrate it into a Streamlit application.
- Display estimated production based on user-provided inputs.

2.4 Glossary

Term	Meaning
AI	Artificial Intelligence
ML	Machine Learning

CSV	Comma-Separated Values
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
R^2	Coefficient of Determination
Random Forest	Ensemble learning algorithm combining multiple decision trees
Streamlit	Python framework for interactive data applications
Joblib	Python library used to save and load model objects

3. Problem Statement

Agricultural production differs across states, districts, seasons and crop types. Understanding production patterns and estimating crop output can be challenging when information is available only in raw tabular form. Manual analysis of records can take time and may not provide an interactive way to explore different combinations of agricultural inputs.

The project addresses this problem by preparing historical crop data, training a machine-learning regression model and providing a web application through which users can enter agricultural details and view an estimated production value.

3.1 Problems Addressed

1. Difficulty analysing large agricultural records manually.
2. Difficulty identifying crop production patterns from tabular data.
3. Lack of a simple interactive interface for trying different input combinations.
4. Need for a data-driven approach to estimate a numerical production value.

4. Dataset and Technologies

4.1 Dataset Description

The project explanation identifies the dataset as crop_production_starter_dataset.csv. It contains 60 records and 7 columns.

Column	Description	Role
State	Name of the Indian state	Input feature
District	Name of the district	Input feature
Crop_Year	Year of cultivation	Input feature
Season	Agricultural season	Input feature
Crop	Name of the crop	Input feature
Area	Area used for cultivation	Input feature
Production	Crop production quantity	Target variable

The target variable is Production. The other six columns are used as input features. State, District, Season and Crop are categorical features, while Crop_Year and Area are numerical features.

4.2 Technologies Used

Technology	Purpose
Python	Main programming language for data processing, model development and application logic.

Pandas	Loading, inspecting and manipulating the CSV dataset.
scikit-learn	Building and evaluating the machine-learning model.
Random Forest Regression	Estimating the numerical crop production target.
Streamlit	Creating the interactive web application.
Joblib	Saving and loading the trained model.

5. Proposed Design and Methodology

5.1 System Architecture

The system follows a sequence from agricultural dataset preparation to production prediction:

Stage	Description
1. Agricultural dataset	CSV containing 60 records and 7 columns.
2. Data loading and inspection	Read the file using Pandas and inspect records, columns and data types.
3. Data preprocessing	Prepare categorical and numerical features and handle unsuitable or missing values where required.
4. Random Forest Regression	Train an ensemble regression model using the prepared features.
5. Model evaluation	Evaluate predictions using MAE, RMSE and R^2.
6. Model persistence	Save the trained model using Joblib.
7. Streamlit application	Collect agricultural inputs and load the saved model.
8. Prediction output	Display the estimated crop production value.

5.2 Step-by-Step Methodology

5. Dataset collection: start with the agricultural crop production CSV file.
6. Data loading: use Pandas read_csv() to load the dataset and inspect it with head() and info().
7. Data preprocessing: inspect data types, prepare categorical and numerical variables, handle missing or unsuitable values where necessary, and prepare the target column.
8. Feature preparation: encode categorical values into numerical representations so they can be used by the model.
9. Model selection: choose Random Forest Regression, an ensemble method that combines multiple decision trees.
10. Model training: train the model using the prepared input features and Production as the target.
11. Model evaluation: calculate MAE, RMSE and R^2 to understand prediction errors and explained variation.
12. Model saving: save the trained model as models/crop_model.joblib and evaluation metrics as models/metrics.json.
13. Application development: build a Streamlit interface that accepts input values, loads the saved model and displays a production estimate.

5.3 Why Random Forest Regression?

Random Forest is an ensemble learning technique. It creates multiple decision trees and combines their outputs, usually by averaging, to produce the final regression prediction. It can learn relationships among multiple input features and a numerical target.

For illustration only, if three trees predict 100, 120 and 110 tonnes, the average prediction is 110 tonnes. This is a simplified example of how regression outputs can be combined.

- Can learn complex relationships in data.
- Combines multiple decision trees.
- Can handle different input types after suitable preprocessing.
- Can reduce the influence of errors from an individual tree.
- Can be used for both regression and classification tasks.

6. Dashboard and User Interface

The AgriPredict application is presented as a Streamlit-based Smart Agriculture Platform. The supplied dashboard screenshot shows a left navigation panel and a main welcome area.

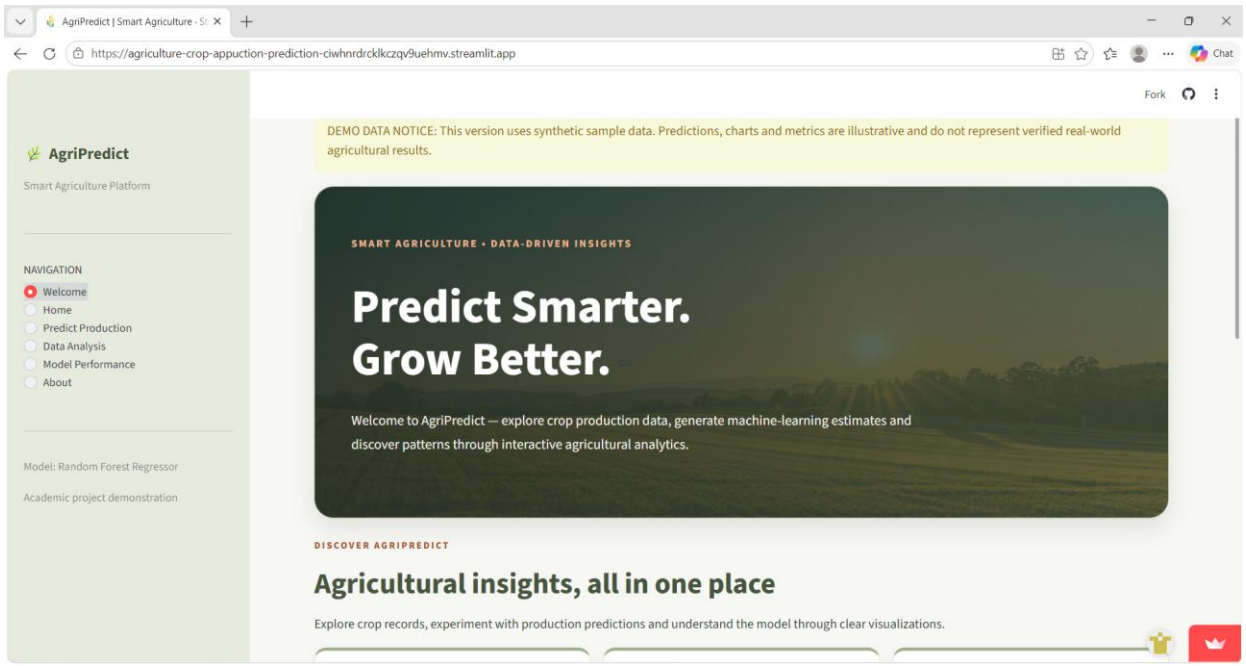


Figure 1. AgriPredict dashboard home screen (screenshot supplied for this report).

6.1 Dashboard Components

Dashboard element	Function shown in the screenshot
AgriPredict branding	Displays the application name and Smart Agriculture Platform label.
Welcome	Landing page selected by default.
Home	Navigation entry for the main application area.
Predict Production	Entry for generating crop production estimates.
Data Analysis	Entry for exploring agricultural data.
Model Performance	Entry for viewing model evaluation information.
About	Entry for project and application information.
Model information	Sidebar identifies the model as Random Forest Regressor.
Academic project notice	Identifies the application as an academic project demonstration.

The screenshot also contains a demo data notice stating that the displayed version uses synthetic sample data and that predictions, charts and metrics are illustrative rather than verified real-world agricultural results. This qualification should remain visible during demonstrations.

6.2 User Workflow

14. Open the AgriPredict Streamlit application in a browser.
15. Choose the relevant navigation section, such as Predict Production.
16. Enter or select the agricultural input values requested by the application.
17. Submit the inputs for processing by the saved Random Forest model.
18. Review the estimated crop production displayed by the application.
19. Use Data Analysis and Model Performance sections to explore available analysis and evaluation information.

7. Model Evaluation and Results

The supplied project explanation records the following evaluation values:

Metric	Recorded value	Interpretation
MAE	108,681.79	Average absolute difference between actual and predicted production.
RMSE	129,101.07	Prediction error metric that gives greater weight to larger errors.
R^2 Score	0.447	The model explains approximately 44.7% of the variation in the evaluation data.

A lower MAE and RMSE generally indicate smaller prediction errors. An R^2 score of 0.447 indicates that there is room to improve the model. It must not be described as 44.7% accuracy.

The project explanation also gives an example prediction of approximately 174,112.52 production units. This is an illustrative model output, not an official agricultural estimate.

7.1 Result Qualification

The dataset contains only 60 records. Therefore, the model is a prototype for learning and demonstration and should not be considered a reliable nationwide agricultural forecasting system. The reported metrics should be interpreted in the context of this limited starter dataset.

8. Testing, Limitations and Future Scope

8.1 Functional Test Plan

Test ID	Test case	Expected outcome
TC-01	Open the Streamlit dashboard	Welcome page and navigation load.
TC-02	Submit valid agricultural inputs	A production estimate is returned.
TC-03	Submit missing or invalid inputs	Application provides validation feedback.
TC-04	Open Data Analysis	Available data analysis content is displayed.
TC-05	Open Model Performance	Recorded model metrics are displayed.
TC-06	Open About	Project information is displayed.

The supplied project material does not include a formal test execution log. The above cases are recommended functional checks; mark them as passed only after running them on the deployed application.

8.2 Limitations

- The starter dataset contains only 60 records.
- The dataset may not represent all Indian states, districts and crops.
- Model performance depends on the quality and coverage of the available data.
- The current feature set does not establish the effect of all real-world factors such as rainfall, soil quality and irrigation.
- The R^2 score indicates scope for improvement.
- Predictions are estimates and are not guaranteed production quantities.
- The dashboard screenshot explicitly identifies the current display as using synthetic sample data for demonstration.

8.3 Future Enhancements

- Collect larger datasets from reliable agricultural sources.
- Add rainfall, temperature and soil information.
- Include irrigation and fertilizer-related information where available.
- Compare Random Forest with XGBoost, Decision Tree and other regression algorithms.
- Improve performance through hyperparameter tuning.
- Add state-wise and crop-wise production graphs.
- Develop more detailed historical trend analysis.
- Deploy the application for wider access.

9. My Learnings

- Using Python and Pandas to load, inspect and prepare structured datasets.
- Understanding categorical and numerical feature preparation.
- Learning the fundamentals of Random Forest Regression.
- Training and evaluating a regression model using MAE, RMSE and R^2.
- Saving and loading trained model files with Joblib.
- Building an interactive data application using Streamlit.
- Understanding how a machine-learning model can be connected to a user interface.
- Documenting model limitations and communicating results responsibly.

9.1 Conclusion

AgriPredict demonstrates how machine learning can be applied to agricultural data analysis. The project uses Python and Pandas for data handling, Random Forest Regression to estimate crop production, and MAE, RMSE and R^2 to evaluate the model. The trained model is saved with Joblib and integrated into a Streamlit web application.

The project provided practical experience in dataset preparation, regression modelling, model evaluation and web application development. Although the current model is based on a small starter dataset, it provides a foundation for a more advanced agricultural prediction system using larger and more reliable data.

10. References

Python documentation: https://docs.python.org/3/

Pandas documentation: https://pandas.pydata.org/docs/

scikit-learn documentation: https://scikit-learn.org/stable/

Streamlit documentation: https://docs.streamlit.io/

Joblib documentation: https://joblib.readthedocs.io/

Project explanation supplied for Prediction of Agriculture Crop Production in India.

Appendix A – Viva Preparation

What is the main aim of the project?

To estimate agricultural crop production using historical agricultural data and machine learning.

Which algorithm is used?

Random Forest Regression.

What is the dataset size?

60 records and 7 columns.

What is the target variable?

Production.

Why is regression used?

Production is a numerical value, so the task is to estimate a quantity.

Why is Streamlit used?

To create an interactive web application using Python.

What is Joblib used for?

To save and load the trained machine-learning model.

What does an R^2 score of 0.447 mean?

The model explains approximately 44.7% of the variation in the evaluation data; it is not 44.7% prediction accuracy.

What is the main limitation?

The dataset contains only 60 records and may not represent the full diversity of Indian agriculture.

What is the future scope?

Use larger datasets, add weather and soil features, compare algorithms and tune model parameters.

Appendix B – Details to Complete Before Submission

- Add USN / registration number if required by the college.
- Add faculty guide and internship mentor names, if applicable.
- Add internship start and end dates if required.
- Insert the actual deployed application URL and source-code repository URL.
- Confirm that the displayed model metrics match the latest evaluation run.
- Run the functional tests and record actual outcomes.